

Τμήμα Μηχανικών Ηλεκτρονικών Υπολογιστών & Πληροφορικής

Προχωρημένα Μαθηματικά Ι (ΕΕΙ_101)

Εργασία Εξαμήνου
ΦΘΙΝΟΠΩΡΙΝΟ ΕΞΑΜΗΝΟ
2024-2025

Φοιτητής:

Γεώργιος Κάρυος

(Αριθμός Φοιτητή: 30738)

Καθηγητής:

Δρ. Μαρία Αλανέλλη

1 Δεκεμβρίου, 2024

(01/12/2024)

(30738)

Εργασία Εξαμήνου: Κάρυος Γιώργος

ΕΣ 7.1) (3) a) $\lim_{x \to 3^-} f(x) = -1$ b) $\lim_{x \to 3^+} f(x) = 3$ c) $\lim_{x \to 3} f(x) = \nexists$

d) $f(3) = 7$

(4) a) $\lim_{x \to 2^-} f(x) = 2$ b) $\lim_{x \to 2^+} f(x) = 0$ c) $\lim_{x \to 2} f(x) = \nexists$ d) $f(2) = 2$

(32) a) η αρχική μάζα του αντικειμένου
b) η μάζα τείνει στην ταχύτητα του φωτός

12ace)

ΕΣ 1.2) a) $\lim_{x \to 2} [f(x)+g(x)] = 0$

b) $\lim_{x \to 0} [f(x)+g(x)] = \nexists$ αφού το δεν υπάρχει

c) $\lim_{x \to 0^+} [f(x)+g(x)] = \lim_{x \to 0^+} (-2+2) = 0$

d) $\lim_{x \to 0^-} [f(x)+g(x)] = \lim_{x \to 0^+} (7+2) = 3$

e) $\lim_{x \to 2} \frac{f(x)}{7+g(x)} = 0$

f) $\lim_{x \to 2} \frac{7+g(x)}{f(x)} = \nexists$ αφού ο παρονομαστής τείνει στο 0

g) $\lim_{x \to 0^+} \sqrt{f(x)} = \nexists$ Δεν υπάρχει η ρίζα

h) $\lim_{x \to 0^+} \sqrt{f(x)} = 7$

8) $\lim_{t\to -2} \frac{t^3+8}{t+2} = \lim_{t\to -2} \frac{(t+2)(t^2-2t+4)}{t+2} = 12$

ΕΣ) 1.3) (30) $\lim_{x\to+\infty} \frac{\sqrt{3x^4+x}}{x^2-8} = \lim_{x\to+\infty} \frac{x^2\sqrt{3+\frac{7}{x^3}}}{x^2(7-\frac{8}{x^2})} = \sqrt{3}$

34) Σωστο 35) λάθος

46)

Εx.S.) 1.5) 1) λαθος αφου το $\lim_{x\to 2} f(x) = \not\exists$

2) b) λάθος αφου $\lim_{x\to 2} f(x) \neq F(2)$

3) b) σωστο & συνεχής

e) λάθος, το $F(3)$ δεν υπάρχει

(10) a) Δεν είναι συνεχής

b) συνεχής

c) Οχι συνεχής

d) συνεχής

(12) $F(x) = \sqrt[3]{x-8}$ Δεν Ξ ζητές ασυνέχειας αφου η F είναι τριωνυμική

(16) $F(x) = \frac{2x+1}{4x^2+4x+5}$ Δεν Ξ ζητές ασυνέχειας, ο παρανομαστής δεν μηδενίζεται

(28) Λαθος αφου δεν γνωρίζουμε αν ορίζει η $\sqrt{F(x)}$

(31) $\lim_{x\to 2^+} x^2+5 = 9$

$\lim_{x\to 2^-} f(x) = 3m+k = 3m+4$ } $3m+4=9$, $\boxed{m=\frac{5}{3}}$

$\lim_{x\to -1^-} f(x) = 4$

$\lim_{x\to -1^+} f(x) = k$ } $\boxed{k=4}$

Ex. 5. 7.6)

(2) Lecture notes in Calculus : (7) a) $\lim_{x\to a}[f(x)-g(x)+2h(x)]$

$= \lim_{x\to a} f(x) - \lim_{x\to a} g(x) + 2\lim_{x\to a} h(x) = 7 - 2 + 0 = -7$

β) $\lim_{x\to a}[f(x)-g(x)h(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)h(x) = 7-(-2)(0) = 7$

γ) $\lim_{x\to a}\left[\frac{f(x)h(x)}{g(x)}\right] = \frac{\lim_{x\to a} f(x)h(x)}{\lim_{x\to a} g(x)} = \frac{0}{-2} = 0$

δ) $\lim_{x\to a}\frac{2f(x)+g(x)}{h(x)} = \frac{2\lim_{x\to a} f(x) + \lim_{x\to a} g(x)}{\lim_{x\to a} h(x)} = \frac{2-2}{0} = \frac{0}{0}$ Απρ

ε) $\lim_{x\to a}[f(x)g(x)]^2 = \left[\lim_{x\to a} f(x)g(x)\right]^2 = 4$

στ) $\lim_{x\to a}\sqrt{\frac{f(x)}{g(x)}} = \sqrt{\lim_{x\to a}\frac{f(x)}{g(x)}} = \sqrt{2}$

9) b) $\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty}\frac{x^4-1}{2x-2} = \lim_{x\to-\infty}\frac{x^4}{2x} = -\infty$

3) Calculus chapter 2

Exs 2.2 (2) $v_1 = \frac{x}{t} = \frac{23}{4}$ m/s $v_2 = \frac{x}{t} = \frac{97}{8}$ m/s

$f'(4) < f'(2) < f'(0) < 0 < f'(-3)$

8) $(-2, 3)$ $m = +4$

$y - 3 = -4(x+2)$

$\Rightarrow y - 3 = -4x - 8$

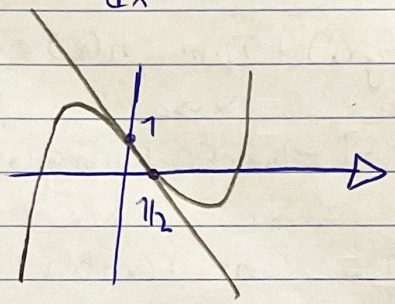
$\Rightarrow y + 4x = -5$

(30) Παθος

35) $y = x^3 - 2x + 7$ $\frac{dy}{dx} = 3x^2 - 2$ $m = \left.\frac{dy}{dx}\right|_{(0,7)} = -2$

$y - 7 = -2x$

$\boxed{y + 2x = 7}$



2.3) (6) $f'(x) = \sqrt{2}$

(8) $y' = \frac{(x^2+7)' \cdot 8}{25} = \frac{2x}{5}$

(12) $f'(x) = -42x^{-7} - \frac{5}{2\sqrt{x}}$

(15) $f'(x) = 3ax^2 + 2bx + c$

(24) $\frac{dy}{dx} = 24x^{23} + 24x^{11} + 24x^7 + 24x^5$

$\left.\frac{dy}{dx}\right|_{x=1} = 96$

(48) $m = -1$ $(2, -2)$

$y + 2 = -1(x - 2)$

$\boxed{y = -x}$

43) a) $y' = -5x^{-6} + 5x^4$

$y'' = 30x^{-7} + 20x^3$

$y''' = 210x^{-8} + 60x^2$

b) $y' = \frac{7}{x^2}$

$y'' = \frac{7}{x^4}$

$y'' = \frac{1}{x^8}$

c) $f'(x) = 3ax^2 + b$

$f''(x) = 6ax$

$f'''(x) = 6a$

46) a) $y' = 16x^3 + 6x^2$

$y'' = 48x^2 + 12x$

$y''' = 96x + 12$

$y''''(0) = 12$

$f^{IIII}(x) = \frac{5040}{x^8} \Rightarrow f^{IIII}(x) = 5040$

2.4)

KARYOZ F
(30738)

~~2.4)~~ 10)

$f'(x) = (x^2+x)(x^2-x) = x^4 - x^3 + x^3 - x^2 = x^4 - x^2$

$f'(x) = 4x^3 - 2x$

50) J ~~Z~~

2.5) 17) Find $f'(x)$ $\Rightarrow f(x) = \frac{\sin x \sec x}{7 + x\tan x}$

$$f'(x) = \frac{(\sin x \sec x)'(7 + x\tan x) - (\sin x \sec x)(7 + x\tan x)'}{(7 + x\tan x)^2}$$

$$= \frac{(\sec\tan x \sin)}{7 + x\tan x} + \frac{\sec x \cos x}{7 + \tan x} - \frac{(\sin x \sec)(\tan x + x\sec^2 x)}{(7 + x\tan^2)}$$

18) $f'(x) = \frac{(x^2+1)\cos x}{3 - \cos x + \csc x}$ $\quad \frac{U}{V}$

$$= \frac{[2x\cot x - (2x^2+1)\csc x](3 - \cos x \csc x) - [(x^2+1)\cos x](\sin x \cos x + \tan x \csc x)}{(3 - \cos\csc x)^2}$$

. . . .

21) $y = \sin x - 3\cos x$

$y' = \sin x + x\cos x + 3\sin x = x\cos + 4\sin x$

$y'' = \cos x - x\sin x + 4\cos x = 5\cos x - x\sin x$

25a) $y = \tan x \quad y' = \sec x$

$\sec 0 = \quad y - y_1 = f(x - x_1) \Rightarrow y = x$

b) $x = \pi/4 \quad \sec \pi/4 = \frac{2}{\sqrt{2}} \quad y = 1(\frac{\pi}{4} + 1)$

$y = \frac{2x}{\sqrt{2}} - \frac{\pi}{2\sqrt{2}} + 1$

c) $x = -\pi/4 \quad (-\frac{\pi}{4}, -1)$

$\tan(-\pi/4) = -1$

$\sec(-\frac{\pi}{4}) = \sqrt{2}$

$\Rightarrow y = \sqrt{2}x + \frac{\sqrt{2}\pi}{4} - 1$

2.6) ~~7)~~ 7) $F(x) = (x^3 + 2x)^{37}$

$F'(x) = 37(x^3+2x)^{36}(3x^2+2)$

12) $F(x) = \sqrt{x^3 - 2x + 5} = (x^3 - 2x + 5)^{1/2}$

$F'(x) = \frac{(3x^2 - 2)}{2\sqrt{x^3 - 2x + 5}}$

28) $y = \sqrt{x}\tan^3(\sqrt{x})$

$y' = (\tan^3(\sqrt{x}))' \cdot \sqrt{x} + \tan^3(\sqrt{x}) \cdot (\sqrt{x})'$

$= 3\tan^2(\sqrt{x})\sec^2\sqrt{x} \cdot \frac{\sqrt{x}}{2\sqrt{x}} + \frac{\tan^3(\sqrt{x})}{2\sqrt{x}}$

$y' = \frac{3\sec^2(\sqrt{x})\tan^2(\sqrt{x})}{2} + \frac{\tan^3(\sqrt{x})}{2\sqrt{x}}$

39) $y = \frac{(2x+3)^3}{(4x^2-7)^8}$

$y = \frac{((2x+3)^3)'(4x^2-7)^8 - (2x+3)^3((4x^2-7)^8)'}{((4x^2-7)^8)^2}$ ...

45) $y = \sec^3(\frac{\pi}{2} - x)$, $x = -\frac{\pi}{2}$

$f(x) = \sec^3(\frac{\pi}{2} - x)$

$f(-\frac{\pi}{2}) = \sec^3(\pi) = -1$ $\quad (-\frac{\pi}{2}, -1)$

$f'(x) = 3\sec^2(\frac{\pi}{2} - x)(-\csc x \cot x)$

$f'(\pi/2) = 0$ $\quad y - y_1 = 0 \Rightarrow \boxed{y = -1}$

(4) CALCULUS

3.2) 2) $y = \ln\frac{x}{3}$

$y' = \frac{1/3}{x/3} = \frac{1}{x}$ $\quad (\frac{x}{3})' = \frac{1}{3}$

$y' = \frac{-1}{\frac{3}{x/3}} = \frac{1}{x}$

$\ln x = x' \cdot \frac{1}{x}$

8) $y = \ln\left|\frac{1+x}{1-x}\right|$

$y' = \frac{(\frac{1}{1-x}) + \frac{1+x}{(1-x)^2}}{\frac{(1+x)}{1-x}}$

ΚΑΡΥΟΣ ΤΣΕΡΤΟΣ

(30738)

4, calculus chapter 3

3,2 2) $y = \ln x_3$

$y' = \frac{1}{x_3} \cdot \frac{1}{3} = \frac{1}{x}$

8) $y = \ln\left|\frac{1+x}{1-x}\right| = \ln|1+x| - \ln|1-x|$

$y = \frac{1}{1+x} + \frac{1}{1-x}$

12) $y = \ln\sqrt{x}$

$y = \frac{1}{2x}$

16) $\ln y = \ln x + 3\ln(\log_2(x^2-2x))$

$y' = x(\log_2(x^2-2x))^3\left[\frac{1}{x} + \frac{6(x-1)}{\ln 2\,(x^2-2x)\log_2(x^2-2x)}\right]$

3,3 7) $f(x) = 5x^3 + x - 7$

$f^{-1}(x) = 5y^3 - y - 7$

$y = \frac{1}{15y^2+1}$

8) $f(x) = \frac{1}{x^2}$

$f^{-1}(x) = \frac{1}{y^2}$

$(f^{-1}(x))' = -\frac{y^3}{2}$

33) $y = \ln x^{\tan x}$

$\ln y = \ln(\ln x)^{\tan x}$

$\ln y = \tan x \ln(\ln x)$

$(\ln x)^{\tan x}\left(\frac{\tan x}{x\ln x} + \sec x \ln(\ln x)\right)$

38) $y = \cos^{-1}\left(\frac{x+1}{2}\right)$

$y' = \frac{1}{2\sqrt{1-\left(\frac{x+1}{2}\right)^2}}$

3,6 8) a) $\lim_{x\to 0} \frac{\sin x}{\tan x} = \lim_{x\to 0} \frac{\sin x}{\frac{\sin}{\cos x}} = 1$

b) $\lim_{x\to 1} \frac{x^2-1}{x^3-1} = \lim_{x\to 1} \frac{(x-1)(x+1)}{(x-1)(x^2+x+1)} = \frac{2}{3}$

exercise set 3.5

(28) 6.0075.

10) $\lim_{t\to 0} \frac{t\cdot e^t}{1-e^t} = \frac{0}{0}$ (Adpoad)

$\lim_{t\to 0}\left(\frac{t\cdot e^t}{1-e^t}\right)' = \lim_{x\to 0} \frac{e^t + te^t}{-e^t} = -1$

26) $\lim_{x\to\pi} (x-\pi)\cos x = 0$

$\lim_{x\to\pi}((x-\pi)\cos x)' = \lim_{x\to\pi} \frac{1}{\sec^2 x} = 1$

32) $\lim_{x\to\infty} e^{\ln\cos x\left(\frac{2}{x}\right)^{x^2}} = \lim_{x\to\infty} \cos\left(\frac{2}{x}\right)^{x^2} =$ (Adpoad)

$\lim_{x\to\infty}\left(e^{\ln\cos x\left(\frac{2}{x}\right)^{x^2}}\right)' = e^{-2}$

5. Calculus Chapter 4

4.1 12) $f'(x) = -(x-0,9)(x-1,1)$

$f'(1) > 0$

16) $f(x) = 5 - 4x - x^2$

$f'(x) = -4 - 2x$

$f''(x) = -2$

a) $f(x)$ aşağıya $(-\infty, -2)$

b) $f(x)$ yukarıya $(-2, +\infty)$

4.2 8) $f(x) = 3x^4 + 12x$

$f'(x) = 12x^3 + 12$

$= 12(x+1)(x^2 - x + 1)$

4.4 9) $f'(x) = 3(x-2)^2$

$f'(x) = 0$ $\quad f(2) = 0$

$3(x-2)^2 = 0$ $\quad f(1) = 1$

$x = 2$ $\quad f(4) = 4$

Ex. Set 4.4

$f'(x) = 3(x-2)^2 \Rightarrow 3(x-2)^2 = 0$

$f'(x) = 0$

$x = 2$

$f(2) = 0$

$f(1) = 1$

$f(4) = 4$

Exercise Set 4.5)

7) $\frac{x}{6}+\frac{y}{8}=1$

$y=8(1-\frac{x}{6})=8-\frac{4x}{3}$

$E_{\text{επιφ}}=xy \Rightarrow x(8-\frac{4x}{3})=8x-\frac{4x^2}{3}$

$\frac{dE_{\text{επιφ}}}{dx}=\frac{d(8x-\frac{4x^2}{3})}{dx}=8-\frac{8x}{3}$

$8-\frac{8x}{3}=0$ για μέγιστο

$8=\frac{8x}{3} \Rightarrow \boxed{x=3} \Rightarrow y=8-\frac{4\cdot 3}{3} \Rightarrow \boxed{y=4}$ W: $x=3$, H: $y=4$

22) $x^2=h=2550$

$h=\frac{2550}{x}$

$C=2\cdot 2x^2=4x^2$

$=4(x\cdot h)\cdot 3=12x\cdot h$

$=12x\left(\frac{2250}{x^2}\right)=\frac{27000}{x}$

$C(x)=4x^2+\frac{27000}{x}$

$C'(x)=8x-\frac{27000}{x^2}=0 \Rightarrow 8x^3=27000$

$x^3=3375$ $\boxed{x=15}$

$h=\frac{2250}{15^2}=\frac{2250}{225}=10$

$x=15$, $y=10$ } ίντσες

→ ΔΕΝ ΒΡΗΚΑ ΤΙΣ ΑΣΚΗΣΕΙΣ 10.8 ΜΕ 10.10 από CALCULUS chapter 10

Εx.Set 7.5) 5) $\frac{1-x^2}{x^3(x^2+2)}$ $\frac{A}{x}+\frac{B}{x^2}+\frac{\Gamma}{x^3}+\frac{\Delta x+E}{x^2+2}=\frac{1-x^2}{x^3(x^2+2)}$

24) $\int\frac{3x^2-x+1}{x^3-x^2}dx=\int\frac{3x^2-x+1}{x^2(x-1)}dx \ldots =\int\left(\frac{3}{x-1}-\frac{1}{x^2}\right)dx$

$=3\ln(x-1)+\frac{1}{x}+c$

Εx.Set 5.3) 17) $\int\sin 7x\,dx=\frac{1}{7}\int 7\sin 7x\,dx$

$=\frac{-\cos 7x}{7}+c$

19) $\int\sec 4x\tan 4x\,dx=\frac{1}{4}\int 4\sec 4x\tan 4x\,dv=\frac{\sec 4x}{4}+c$

Ex. Set 8.2) 8) $\frac{dy}{dx} - (1+x)(1+y^2) = 0$

$\frac{dy}{dx} = (1+x)(1+y^2) \Rightarrow \int \frac{1}{1+y^2} dy = \int 1+x\,dx$

$\arctan y = x + \frac{x^2}{2} + c \Rightarrow y = \tan\left(x + \frac{x^2}{2} + c\right)$

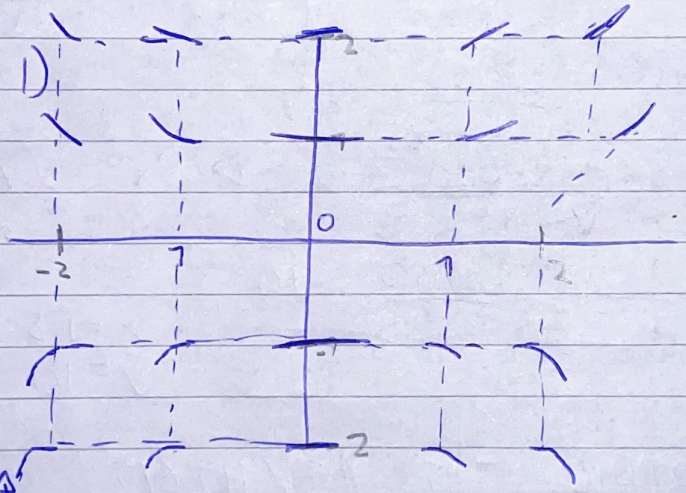
25) $\frac{dy}{dx} = -4xy^2 \quad \int \frac{1}{y^2} dy = \int -4x\,dx \Rightarrow \boxed{y = \frac{1}{2x^2 - c}}$

40) $N(t) = N_0 e^{rt}$ r: μεταβολή αυξησης

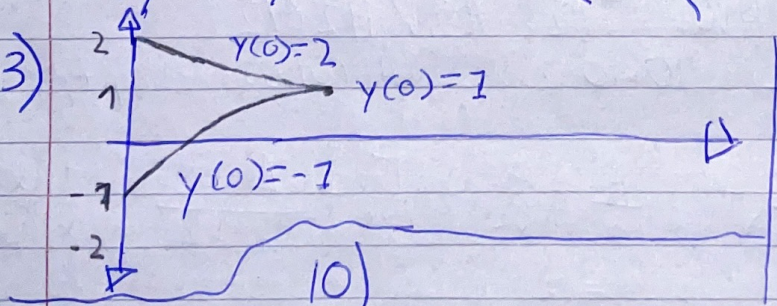
$N(t) = 3N_0$

$3N_0 = N_0 e^{rT} \Rightarrow 3 = e^{rT} \Rightarrow \ln 3 = rT \Rightarrow \boxed{T = \frac{\ln 3}{r}}$

Ex. Set 8.3) 1)



3)



5) $y \to 1$

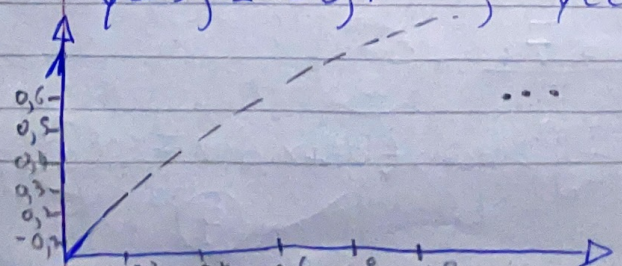
$x \to +\infty$

10)

$\frac{dy}{dt} = e^{-y}, y(0) = 0, 0 \le t \le 1, \Delta t = 0$

$y_{n+1} = y_n + \Delta t \cdot F(t_n, y_n) \Rightarrow f(t,y) = e^{-y}$

$y(0), \Delta t = 0,1 \ldots, y(t)$ από $t=0$ μέχρι $t=1$



... 10 βήματα στο περίπου

Ex. Set 8.4) 13) σωστό

16) $y' = 2y - x$ $\quad y(7)=7$, $y(0)=-7$, $y(-7)=\frac{dy}{dx}=2y-x$

$\frac{dy}{dx} - 2y = -x \Rightarrow e^{\int -2x\,dx} = e^{-2x}$

$e^{-2x}\frac{dy}{dx} - 2e^{-2x} = -2e^{-2x}$ $\quad \boxed{y=-xe^{-2x}}$ $\quad \frac{d}{dx}(ye^{-2x}) = -xe^{-2x}$

$\int -xe^{-2x}dx = -\left(x\cdot\frac{-e^{-2x}}{2}\right) + \int \frac{1}{2}\cdot e^{-2x}dx$

$= \frac{xe^{-2x}}{2} - \frac{1}{4}e^{-2x} + c \Rightarrow y = \frac{x}{2} - \frac{1}{4} + Ce^{2x}$

- $y(7)=1 \Rightarrow 7 = \frac{7}{2} - \frac{1}{4} + Ce^2$

  $7 = \frac{7}{4} + Ce^2 \Rightarrow C = \frac{3}{4e^2}$ $\quad y = \frac{x}{2} - \frac{7}{4} + \frac{3}{4e^2}e^2x$

- $y(0)=-7$ $\quad -7 = 0 - \frac{7}{4} + c \Rightarrow c = -7 + \frac{7}{4}$

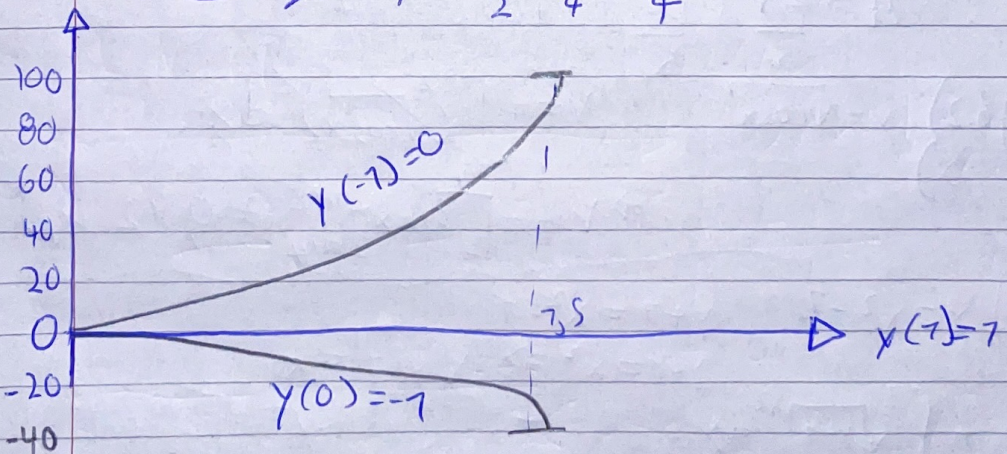
  $c = -3/4$

  $y = \frac{x}{2} - \frac{7}{4} - \frac{3}{4}e^{2x}$

- $y(-7)=0$

  $0 = -\frac{7}{2} - \frac{7}{4} + Ce^{-2} \Rightarrow C = \frac{3}{4}e^2$

  $y = \frac{x}{2} - \frac{7}{4} + \frac{3}{4}e^2 e^{2x}$



ΔΕΝ ΒΡΗΚΑ) ⇒ Ex. Set 6.7) 9) $y=\cos 2x$, $y=0$ $\quad x=\frac{\pi}{4}$ $\quad x=\frac{\pi}{2}$

TH 8.5

$\int_{\pi/4}^{\pi/2} -\cos 2x\,dx = \frac{1}{2}\int_{\pi/4}^{\pi/2} -2\cos 2x\,dx = \frac{1}{2}\left[-\sin 2x\right]_{\pi/4}^{\pi/2} = \frac{1}{2}$

12) $x^2 = y$

$x = y-2 \Rightarrow (y-2)^2 = y \Rightarrow (y-4)(y-7) = 0 \Rightarrow y_1 = 4$ $\Rightarrow y_2 = 7$

$x_1 = 4-2 = 2$

$x_2 = 7-2 \Rightarrow x_2 = -7$ $\int_{-7}^{2} ((x+2) - x^2)\,dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-7}^{2} = \frac{9}{2}$

17) $y = 2 + |x-7|$

$y = -\frac{7}{5}x + 7$ σημεία τομής: $(-5, 8)$, $(5, 6)$

$A = \int_{-5}^{7} \left(-\frac{x}{5} + 7 - (2 + |x-7|)\right) dx + \int_{7}^{5} \left(-\frac{x}{5} + 7 - (2 + |x-7|)\right) dx$

για $x < 7$

$|x-7| = -(x-7) = -x+7 \Rightarrow -\frac{x}{5} + 5 + x - 7 = \frac{4x}{5} + 4$

για $x \geq 7$: $(x-7) = x-7 \Rightarrow -\frac{x}{5} + 5 - (x-7) = \frac{-6x}{5} + 6$

$\int_{-5}^{7} \left(\frac{4x}{5} + 4\right) dx = \left[\frac{2x^2}{5} + 4x\right]_{-5}^{7} = \frac{72}{5}$

$\int_{7}^{5} \left(-\frac{6x}{5} + 6\right) dx = \left[-\frac{3x^2}{5} + 6x\right]_{7}^{5} = \frac{48}{5}$

$A = \frac{72}{5} + \frac{48}{5} = 120/5 = 24$

Ex. Set) 6.2) 5) $y = \sqrt{\cos x}$ $V = \pi \int_{\pi/4}^{\pi/2} (\sqrt{\cos x})^2\,dx$

$\Rightarrow V = \pi \left[\sin x\right]_{\pi/4}^{\pi/2} = 0{,}92$ κμ

8) $y = x^2 - 7$ $y+7 = x^2 - 7$

$y+7 = x^2 \Rightarrow \sqrt{y+7} = x$

$\pi \int_{-7}^{3} \left(\sqrt{y+7}\right)^2 dy$ $\pi \int_{-7}^{3} (y+7)\,dy = \pi \left[\frac{y^2}{2} + y\right]_{-7}^{3} = 8\pi$ κμ

Ex Set: 6.3

9) $y = 2x - 7$, $y = -2x + 3$, $x = 2$

$\frac{y+7}{2}$, $y\frac{-3}{2} = x$

$2x - 7 = -2x + 3 \Rightarrow 4x - 4 = 0 \Rightarrow 4(x-7) = 0$ $\boxed{x = 7}$

$y = 2 \cdot 7 - 7 = 7$ $(7, 7)$

$y = 2 \cdot 2 - 7 = 3$

$y = -2 \cdot 2 + 3 = -7$ ...

... 9) $\Rightarrow V = \pi \int_{-7}^{1} 2^2 - \left(\frac{y-3}{-2}\right)^2 dy + \pi \int_{1}^{3} 2^2 - \left(\frac{y+7}{2}\right)^2 dy$

V_1 $\qquad$ V_2

για V_1: $\left(\frac{y-3}{2}\right)^2 = \frac{(y-3)^2}{4} \Rightarrow V_1 = \pi \int_{-7}^{1} 4 - \frac{(y-3)^2}{4} dy$

$\Rightarrow V_1 = \pi \left[4y - \frac{(y-3)^3}{12}\right]_{-7}^{1} = \frac{10\pi}{3}$ κμ

για V_2: $V_2 = \pi \int_{1}^{3} 2^2 - \left(\frac{y+7}{2}\right)^2 dy = \left(\frac{y+7}{2}\right)^2 = \left(\frac{y+7}{4}\right)^2$

$V_2 = \pi \left[4y - \frac{(y+7)^3}{12}\right]_{1}^{3}$ $V_2 = \pi \left[\left(4\cdot 3 - \frac{(3+7)^3}{12}\right) - \left(4\cdot 1 - \frac{(1+7)^3}{12}\right)\right] \Rightarrow V_2 = \frac{10\pi}{3}$ κμ

$V = V_1 + V_2 = \frac{10\pi}{3} + \frac{10\pi}{3} = \frac{20\pi}{3}$ κμ

6.4) 4) $x = 1/3 (y^2+2)^{3/2}$ $\qquad$ μηκος $= \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

$x = 1/3 (y^2+2)^{3/3}$

$\frac{dx}{dy} = \frac{1}{3} \cdot \frac{3}{2} (y^2+2)^{1/2} \cdot 2y \Rightarrow \frac{dx}{dy} = y(y^2+2)^{1/2}$

$\left(\frac{dx}{dy}\right)^2 = y^2(y^2+2) \Rightarrow 1 + y^2(y^2+2) = 1 + y^4 + 2y^2$

$\Rightarrow (y^2+1)^2$

$= \int_0^1 y^2 + 1 \, dy = \left[\frac{y^3}{3} + y\right]_0^1 = \frac{4}{3}$ μ

6.5) 3) $y = \sqrt{4-x^2}$, $-1 \le x \le 1$ $\qquad$ $S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

$a = -1$
$b = 1$ $\qquad \frac{dy}{dx} = \frac{1}{2\sqrt{4-x^2}} \cdot (-2x) = \frac{-x}{\sqrt{4-x^2}}$

$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(\frac{-x}{\sqrt{4-x^2}}\right)^2 = 1 + \frac{x^2}{4-x^2}$

$= \frac{4 - x^2 + x^2}{4-x^2} = \frac{4}{4-x^2}$

$\dots \sqrt{\frac{4}{4-x^2}} = \frac{2}{\sqrt{4-x^2}}$ $\quad S = \int_{-7}^{7} 2\pi y \cdot \frac{2}{\sqrt{4-x^2}}\, dx$

$S = \int_{-7}^{7} 7\, dx = 4\pi [x]_{-7}^{7} = 4\pi 2 = 8\pi$

9) $y = \sqrt{x} - \frac{7}{3}x^{3/2}$, $7 \le x \le 3$,

$\frac{dx}{dx} = \frac{7}{2\sqrt{x}} - \frac{\sqrt{x}}{2}$

$7 + \left(\frac{dx}{dx}\right)^2 = 7 + \left(\frac{7}{2\sqrt{x}} - \frac{\sqrt{x}}{2}\right)^2$

$S = \int_{7}^{3} 2\pi \left(\sqrt{x} - \frac{7}{3}x^{3/2}\right) \sqrt{7 + \left(\frac{7}{2\sqrt{x}} - \frac{\sqrt{x}}{2}\right)^2}\, dx$

$\Rightarrow S = \frac{16\pi}{9} \approx 5,85$

Εx. Sct) 6.6) $F = 100N$ όταν $x = 0,2m$

$x = 0,2m$

$h = \frac{F}{x} = \frac{100N}{0,2m} = 500 N/m$ $\quad W = \frac{7}{2} k x^2 = \frac{7}{2} \cdot 500 \cdot 0,8^2$

$W = \frac{7}{2} \cdot 250 \cdot 0,64$

$\Rightarrow \boxed{W = 160 J}$

Σημειώσεις Διαλέξεων στο Nογισμο

ΚΕΦ. 6:

6.6) $\int_0^4 \frac{7}{(x-7)^{7/3}}\, dx$ $\quad 0 = x - 7$ $\quad du = dx$ $\quad \int_{-7}^{3} \frac{7}{\sqrt[3]{u}} = \left[\frac{3u^{2/3}}{2}\right]_{-7}^{3}$

$\left(\frac{3 \cdot (3)^{2/3}}{2}\right) - \left(\frac{3 \cdot (-7)^{2/3}}{2}\right) = 7,62$

6.7) $x_0 = 2$, $x_n = 7$, $h = 0,5$, $n = 10$

$A \approx \frac{0,5}{2}(3,5 + 2)(6,2) + 2 \cdot 7,2 + 2 \cdot 6,8 + 2 \cdot 5,7 + 2 \cdot 5,0 + 2 \cdot 6,2 + 2 \cdot 8,6 + 2\, 11,6 + 2 \cdot 10 + 8,5)$

$A \approx \frac{7}{4}(145,8) \Rightarrow \boxed{A = 36,45}$

Γεώργιος ΚΑΡΥΟΣ (30738)

Ex. Set 4.9)

1)a) $\sum_{n=0}^{\infty} \left(\frac{n!+2}{n!}\right) \frac{7}{3^n}$

$\frac{n!+2}{n!} = 7 + \frac{2}{n!}$ $\quad \sum_{n=0}^{\infty} \left(7 + \frac{2}{n!}\right) \frac{7}{3}^n = \sum_{n=0}^{\infty} \frac{7}{3^n} + 2\sum_{n=0}^{\infty} \frac{7}{n! 3^n}$

$\sum_{n=0}^{\infty} \frac{7}{n! 3^n} = e^{7/3}$

β) $\frac{7}{7\cdot 4} + \frac{7}{2\cdot 5} + \frac{7}{3\cdot 6} + \dots \quad \sum_{n=7}^{\infty} \frac{7}{n(n+3)}$

$\Rightarrow \frac{7}{n(n+3)} = \frac{A}{n} + \frac{B}{n+3}$

$7 = A(n+3) + Bn \Rightarrow 7 = (A+B)n + 3A$

για n^1: $A + B = 0$

για σταθερά: $3A = 7$

$A = \frac{7}{3} \Rightarrow B = -\frac{7}{3}$ $\quad \frac{7}{n(n+3)} = \frac{7}{3n} - \frac{7}{3(n+3)}$

$\sum_{n=7}^{\infty} \frac{1}{n(n+3)} = \sum_{n=7}^{\infty} \left(\frac{7}{3n} - \frac{7}{3(n+3)}\right)$

$= \frac{7}{3} \sum_{n=7}^{\infty} \left(\frac{7}{n} - \frac{7}{n+3}\right) = \frac{7}{3}\left(\left(1 - \frac{7}{4}\right) + \left(\frac{7}{2} - \frac{7}{5}\right) + \dots\right)$

$\Rightarrow \frac{7}{3}\left(7 + \frac{7}{2} + \frac{7}{3} - \lim_{n\to\infty}\left(\frac{7}{n+7} + \frac{7}{n+2} + \frac{7}{n+3}\right)\right)$

$\Rightarrow \frac{7}{3}\left(7 + \frac{7}{2} + \frac{7}{3}\right) = \frac{7}{3} \cdot \frac{11}{6} = \frac{11}{18}$

γ) $\frac{\left(7 + 3 + \frac{3^2}{3!} - \frac{3^3}{3!} + \dots\right)\left(7 + \frac{7}{5} + \frac{7}{5^2} + \dots\right)}{2 - \frac{2^2}{2!} + \frac{2^3}{3!} - \frac{2^4}{4!} + \dots}$

$e^{-x} = 7 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$

$\Rightarrow 7 - 3 + \frac{3^2}{2!} - \frac{3^3}{3!} + \dots = e^{-3}$ $\quad \sum_{n=0}^{\infty} \frac{7}{5^n} = \frac{7}{7 - 7/5} = 5/4$

$e^{-x} = 7 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots = 2\left(7 - 2 + \frac{2^2}{2!} - \frac{2^3}{3!} + \dots\right) = 2e^{-2}$

$\Rightarrow \frac{5 \cdot e^{-3}}{4 \cdot 2e^{-2}} = \frac{5e^{-3}}{8e^{-2}} = \boxed{\frac{5}{8e}}$

Σημειώσεις διαλέξεων στο Λογισμό

Κεφ. 5:

5.5) 2α) $\int x^n \cos(ax)\,dx$

$u = x^n$	$du = nx^{n-1}dx$

$dv = \cos(ax)\,dx$

$v = \frac{\sin(ax)}{a}$ $\qquad \int u\,dv = uv - \int v\,dv$

$$\Rightarrow \int x^n \cos(ax)\,dx = x^n \frac{\sin(ax)}{a} - \int \frac{\sin(ax)}{a} \cdot nx^{n-1}\,dx$$

$$= \frac{x^n \sin(ax)}{a} - \frac{n}{a}\int x^{n-1}\sin(ax)\,dx$$

$u = x^{n-1} \Rightarrow du = (n-1)x^{n-2}\,dx$

$dv = \sin(ax)\,dx \Rightarrow v = -\frac{\cos(ax)}{a}$

$$\int x^{n-1}\sin(ax)\,dx = -\frac{x^{n-1}\cos(ax)}{a} + \frac{n-1}{a}\int x^{n-2}\cos(ax)\,dx$$

$$\int x^n \cos(ax)\,dx = \frac{x^n \sin(ax)}{a} - \frac{n}{a}\left(-\frac{x^{n-1}\cos(ax)}{a} + \frac{n-1}{a}\int x^{n-2}\cos(ax)\,dx\right)$$

$$\int x^n \cos(ax)\,dx = \frac{x^n \sin(ax)}{a} + \frac{n}{a^2}x^n \cos(ax) - \frac{n(n-1)}{a^2}\int x^{n-2}\cos(ax)\,dx$$

$$\Rightarrow \frac{x^n \sin(ax)}{a} + \frac{n}{a^2}x^{n-1}\cos(ax) - \frac{n(n-1)}{a^2}\int x^{n-2}\cos(ax)\,dx$$

β) $\int (x^3+2)\cos(6x)\,dx = C$

$$= \frac{(x^3+2)\sin(6x)}{6} - \frac{x\sin(6x)}{36} + \frac{x^2\cos(6x)}{12} - \frac{\cos(6x)}{216} + C$$

3α) $\int (\ln x)^n\,dx$ $\qquad$ θέτω $u = (\ln x)^n$ $\quad du = n(\ln x)^{n-1}\cdot\frac{1}{x}\,dx$

$dv = dx \Rightarrow v = x$

$$\int u\,dv = uv - \int v\,du$$

$$\int (\ln x)^n\,dx = x(\ln x)^n - \int x\,n(\ln x)^{n-1}\cdot\frac{1}{x}\,dx$$

3α)... $\boxed{\int (\ln x)^n dx = x(\ln x)^n - n\int (\ln x)^{n-1} dx}$

β) $\int (\ln x)^4 dx = x(\ln x)^4 - 4\int (\ln x)^3 dx$

$\int (\ln x)^3 dx = x(\ln x)^3 - 3\int (\ln x)^2 dx$

$\int (\ln x)^2 dx = x(\ln x)^2 - 2\int \ln x\, dx$

$\int \ln x\, dx = x\ln x - x$

$\int (\ln x)^4 dx = x(\ln x)^4 - 4(x(\ln x)^3 - 3(x(\ln x)^2 - 2(x\ln x - x))) + c$

$= x(\ln x)^4 - 4(x(\ln x))^3 - 3(x(\ln x)^2 - 2x\ln x + 2x)) + c$

$= x(\ln x)^4 - 4x(\ln x)^3 + 12x(\ln x)^2 - 24x\ln x + 24x + c$

Ex. Set 5.7

$\int \frac{1}{1+\sin x+\cos x} dx = \int \frac{\sec^2(\frac{x}{2})}{2(\tan(\frac{x}{2})+1)} dx$

$u = \tan(\frac{x}{2}) + 1 \Rightarrow du = \frac{\sec^2(\frac{x}{2})}{2} dx$

$= \int \frac{1}{u} du = \ln u \Rightarrow \ln|\tan(\frac{x}{2})+1| + c$

$\Rightarrow \ln\left|\frac{\sin x}{\cos x + 1} + 1\right| + c$

Τέλος Εφ. Εξαμήνου

Γεώργιος ΚΑΡΥΟΣ (30738)

2ε 17